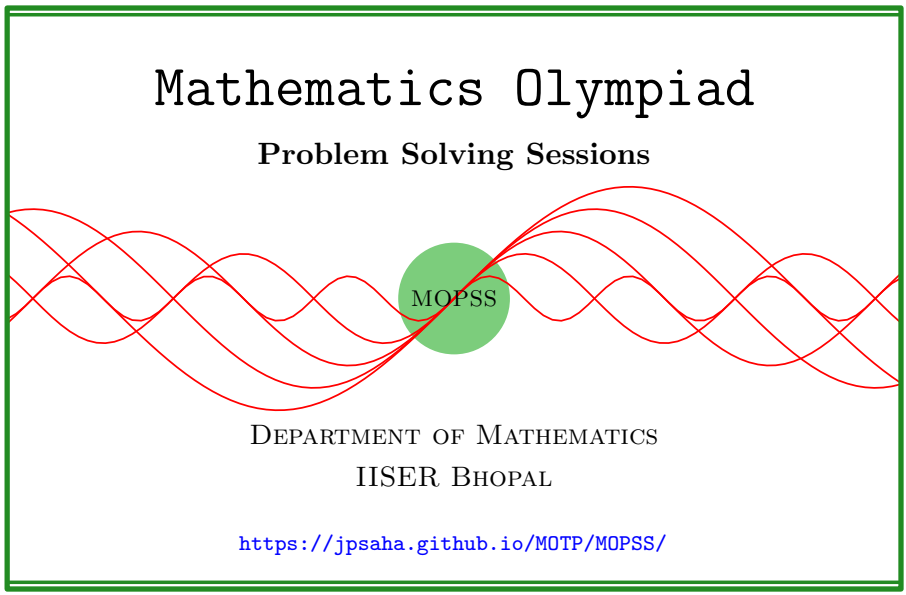
Size of the roots

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

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§1 Size of the roots

Some of the following problems are from [Goy21].

Example 1. Let $f(x)$ and $g(x)$ be nonconstant polynomials with real coefficients such that $f(x^2+x+1) = f(x)g(x)$. Show that $f(x)$ has even degree.

> **Walkthrough** — If the polynomial $f(x)$ admits a real root α, then note that $\alpha^2+\alpha+1$ is also a real root of $f(x)$ and $\alpha^2+\alpha+1 > \alpha$.

Solution 1. If the polynomial $f(x)$ admits a real root α, then note that $\alpha^2+\alpha+1$ is also a real root of $f(x)$ and $\alpha^2+\alpha+1 > \alpha$. Hence, if $f(x)$ possesses a real root, it would have real roots of arbitrarily large absolute value. Since $f(x)$ is nonconstant, it follows that $f(x)$ has no real roots. Since $f(x)$ is a polynomial with real coefficients, it follows that $f(x)$ is of even degree. ■

Exercise 2 (Bulgaria National Olympiad 2000 Day 2 P4,). Find all polynomials $P(x)$ with real coefficients such that

$$P(x)P(x+1) = P(x^2) \quad \text{for all } x \in \mathbb{R}.$$

> **Walkthrough** —
>
> **(a)**

Solution 2.

■

Exercise 3 (Moldova National Olympiad 2001 Grade 12 Day 1 P3,). Find all polynomials $P(x)$ with real coefficieints such that

$$P\left(x^2\right) = P(x)P(x-1)$$

for all $x \in \mathbb{R}$.

Walkthrough —

(a)

Solution 3. Let $P(x)$ be a polynomial with real coefficients satisfying the given functional equation. Note that the polynomials $P(x^2)$, $P(x)P(x-1)$ are equal.

Claim — Any root of $P(x)$ is equal to 0 or is a root of unity.

Proof of the Claim. Let α be a root of $P(x)$ in $\mathbb{C}$. It follows that α^2 is a root of $P(x)$. Since $P(x)$ has finitely many roots, and it vanishes at α^{2^n} for any positive integer n, we conclude that $\alpha = 0$ or α is a root of unity. □

Let β be a root of $P(x)$ in $\mathbb{C}$. Note that $(\beta+1)^2$ is a root of $P(x)$. This shows that if $P(x)$ has real root, then it has a nonnegative real root. Further, if $P(x)$ admits a nonnegative real root, then it admits infinitely many nonnegative real roots, which is impossible. This implies that $P(x)$ admits no real roots.

By the above claim, it follows that $(\beta+1)^2$ is equal to 0 or is a root of unity. Since $P(x)$ has no real roots, we obtain that $\beta+1$ is a root of unity. If $\beta+1=0$, then $\beta=-1$, which is impossible since $P(x)$ has no real roots. Otherwise, $\beta+1$ is a root of unity, which gives $(\beta+1)(\overline{\beta}+1)=1$, implying that $\beta+\overline{\beta}=-1$. Hence, β and $\overline{\beta}$ are the primitive third roots of unity.

Since $P(x)$ has real coefficients, it follows that $\overline{\beta}$ is also a root of $P(x)$. Since $P(x)$ has real coefficients, it follows that the multiplicity of the roots β and $\overline{\beta}$ of $P(x)$ are equal. Using that $P(x)$ has no real roots, we conclude that

$$P(x) = c(x^2+x+1)^k$$

for some $c \in \mathbb{R}$ and nonnegative integer k.

Using that $P(x^2)$ and $P(x)P(x-1)$ are equal polynomials, we obtain that $c^2 = c$, implying that $c=0$ or $c=1$.

Note that

$$(x^2+x+1)^k((x-1)^2+(x-1)+1)^k = (x^2+x+1)^k(x^2-x+1)^k = ((x^2)^2+x^2+1)^k$$

holds.

Therefore, the zero polynomial and the polynomials of the form $(x^2+x+1)^k$, where k is a nonnegative integer, are precisely all the polynomials satisfying the given condition. ■

Example 4. Find all polynomials $P(x)$ (with complex coefficients) satisfying

$$P(x)P(x+2) = P(x^2).$$

Summary — Note that if α is a root of P, then so are α^2 and $(\alpha-2)^2$. Considering absolute values, show that P cannot have a root other than 1. Conclude that $P(x) = c(x-1)^n$.

Walkthrough —

(a)

Solution 4. Let $P(x)$ be a polynomial with complex coefficients satisfying the given condition.

Claim — If P has a root $\alpha \neq 1$, then P has a root which has absolute larger than the absolute value of α.

Proof of the Claim. Note that if $|\alpha| \leq 1$, then

$$\begin{aligned} |(\alpha-2)^2| &\geq (2-|\alpha|)^2 \\ &\geq 1 \\ &\geq |\alpha|. \end{aligned}$$

Moreover, if $|(\alpha-2)^2| = |\alpha|$ holds, then $|\alpha| = 1$ and $\alpha = 2r$ for some real number $r \geq 0$, and hence, $\alpha = 1$. This shows that if $\alpha \neq 1$, and $|\alpha| \leq 1$, then $|(\alpha-2)^2| > |\alpha|$ holds. Also note that if $|\alpha| > 1$, then $|\alpha^2| > |\alpha|$ holds. Since $\alpha^2, (\alpha-2)^2$ are also roots of P, the Claim follows. □

If P has a root other than 1, then applying the above Claim to a root of P having the largest absolute value yields a contradiction. Hence, P is a constant polynomial, or the only root of P is equal to 1.

If P is a constant, then P is equal to 0 or 1. If P is a nonconstant polynomial, then P is equal to $c(x-1)^n$ for some $c \in \mathbb{C} \setminus \{0\}$, and an integer $n \geq 1$. Using the hypothesis, it follows that $c = 1$.

Observing that the polynomials $0, 1, (x-1)^n$ satisfy the given condition, we conclude that there are all the required polynomials. ■

Exercise 5 (Problem 4.12, Putnam training problems by Miguel A. Lerma)**.** Does there exist a polynomial $f(x)$ satisfying

$$xf(x-1) = (x+1)f(x)?$$

Walkthrough —

(a)

Solution 5. Note that the zero polynomial is the only constant polynomial which satisfies the given condition.

Suppose there exists a nonconstant polynomial $f(x)$ satisfying the given condition.

> **Claim —** Any root of $f(x)$ is an integer.

Proof of the Claim. On the contrary, let us assume that α is root of $f(x)$, and that α is not an integer. Note that for any root of β of $f(x)$ in $\mathbb{C} \setminus \mathbb{Z}$, the element $\beta - 1$ is also a root of $f(x)$ lying in $\mathbb{C} \setminus \mathbb{Z}$. It follows that for any integer $k \geq 1$, the element $\alpha - k$ is also a root of $f(x)$. Taking k to be an integer larger than $|\alpha| + \sum_{\gamma \in \mathbb{C}, f(\gamma)=0} |\gamma|$, we obtain

$$|\alpha - k| \geq k - |\alpha| > |\gamma|,$$

and hence $f(x)$ has a root having absolute value larger than that of any of its roots, which is impossible. □

A similar argument can be used to prove that $f(x)$ does not vanish at any negative integer. Moreover, noting that if $f(x)$ vanishes at an integer $n \geq -1$, then $f(x)$ also vanishes at $n+1 \geq -1$, it can be proved using a similar argument that the roots of $f(x)$ are ≤ -2. This contradicts the assumption that there exists a nonconstant polynomial satisfying the given condition. Consequently, the zero polynomial is the only polynomial that satisfies the given condition. ■

Example 6 (Problem 2 of the Problem session for October 28, Fall 2020, Putnam Club)**.** Find all polynomials $P(x)$ satisfying

$$xP(x-1) = (x-20)P(x).$$

Solution 6. Note that the zero polynomial is the only constant polynomial which satisfies the given condition.

Suppose $P(x)$ is a nonconstant polynomial satisfying the given condition.

> **Claim —** Any root of $P(x)$ is an integer.

Proof of the Claim. On the contrary, let us assume that α is root of $P(x)$, and that α is not an integer. Note that for any root of β of $P(x)$ in $\mathbb{C} \setminus \mathbb{Z}$, the element $\beta - 1$ is also a root of $P(x)$ lying in $\mathbb{C} \setminus \mathbb{Z}$. It follows that for any integer $k \geq 1$, the element $\alpha - k$ is also a root of $P(x)$. Taking k to be an integer larger than $|\alpha| + \sum_{\gamma \in \mathbb{C}, P(\gamma)=0} |\gamma|$, we obtain

$$|\alpha - k| \geq k - |\alpha| > |\gamma|,$$

and hence $P(x)$ has a root having absolute value larger than that of any of its roots, which is impossible. □

A similar argument can be used to prove that no root of $P(x)$ is ≤ -1 or ≥ 20. In other words, the roots of $P(x)$ are among $0, 1, 2, \ldots, 19$. The given condition implies that if $0 \leq i \leq 19$ is a root of $P(x)$, then all of the integers $0, 1, 2, \ldots, 19$ are roots of $P(x)$. Hence, $P(x)$ is equal to $cx^{n_0}(x-1)^{n_1}\ldots(x-19)^{n_{19}}$ for some $c \in \mathbb{C}$, and positive integers $n_0, n_1, \ldots, n_{19}$. Using the condition $xP(x-1) = (x-20)P(x)$, it follows that the integers $n_0, \ldots, n_{19}$ are equal to 1. This shows that $P(x)$ is equal to $cx(x-1)(x-2)\cdots(x-20)$. Moreover, any polynomial of this form satisfies the given condition. Hence, the polynomials satisfying the given condition are precisely of the above form. ■

Exercise 7 (INMO 2018 P4,). Find all polynomials $P(x)$ with real coefficients such that $P(x^2+x+1)$ divides $P(x^3-1)$.

Walkthrough —

(a) Show that if α is a root of $P(x)$, then $P(x)$ vanishes at $(\beta_1 - 1)\alpha$ and $(\beta_2 - 1)\alpha$, where β_1, β_2 are the roots of $x^2 + x + 1 = \alpha$.

(b) If α is nonzero, then show that one of $(\beta_1 - 1)\alpha$ and $(\beta_2 - 1)\alpha$ is larger than α in absolute value.

Solution 7. Let us establish the following claim.

Claim — Let α denote a nonzero root of $P(x)$ in $\mathbb{C}$. Then for some $\beta \in \mathbb{C}$ satisfying $\beta^2 + \beta + 1 = \alpha$, the element $(\beta - 1)\alpha$ is a root of $P(x)$ and is larger than α in absolute value.

Proof of the Claim. Let β_1, β_2 denote the roots of $x^2 + x + 1 = \alpha$ in $\mathbb{C}$. Since $P(x^2+x+1)$ divides the polynomial $P(x^3-1)$, it follows that for any $1 \leq i \leq 2$,

$$\beta_i^3 - 1 = (\beta_i - 1)(\beta_i^2 + \beta_i + 1) = (\beta_i - 1)\alpha$$

is a root of $P(x)$. Noting that

$$\begin{aligned} |\beta_1 - 1| + |\beta_2 - 1| &\geq |\beta_1 + \beta_2 - 2| \\ &= |-1-2| \\ &> 2, \end{aligned}$$

we obtain that at least one of $\beta_1 - 1, \beta_2 - 1$ has absolute value larger than 1. Consequently, $|(\beta_i - 1)\alpha| > |\alpha|$ holds for some $1 \leq i \leq 2$. This proves the Claim. □

If $P(x)$ has a nonzero root in $\mathbb{C}$, then applying the above Claim to a root of $P(x)$ of largest absolute value, we would obtain a contradiction. Hence, $P(x)$ is equal to cx^n for some $c \in \mathbb{R}$ and an integer $n \geq 0$. Moreover, any polynomial of this form satisfies the given condition. Consequently, these are all the polynomials satisfying the given condition. ■

The content posted here and at this blog by Evan Chen are quite useful.

Remark. Note that proving $|(\beta_1 - 1)(\beta_2 - 1)| > 1$, in order to conclude that at least one of $\beta_1 - 1, \beta_2 - 1$ has absolute value larger than 1, does not seem to work. Moreover, $|(\beta_1 - 1)(\beta_2 - 1)|$ is smaller than $\frac{1}{2}(|\beta_1 - 1| + |\beta_2 - 1|)$. **Unsurprisingly**, a lower bound for the bigger quantity can easily be obtained.

Exercise 8 (IMOSL 1979, Bulgaria)**.** Find all polynomials $f(x)$ with real coefficients satisfying

$$f(x)f(2x^2) = f(2x^3 + x).$$

Walkthrough —

(a)

Solution 8. Note that $0, 1$ are the only constant polynomials satisfying the given condition.

Let $f(x)$ be a nonconstant polynomial with real coefficients satisfying the given condition.

Claim — The roots of $f(x)$ in $\mathbb{C}$ are of absolute value at most 1.

Proof of the Claim. If α is a root of $f(x)$ in $\mathbb{C}$, then $2\alpha^3 + \alpha$ is a root of $f(x)$. If $|\alpha| > 1$, then

$$|2\alpha^3 + \alpha| \geq 2|\alpha|^3 - |\alpha| > |\alpha|$$

holds. If some root of $f(x)$ has absolute value larger than 1, then taking α to be a root of $f(x)$ with largest absolute value, we would obtain a contradiction. This proves the Claim. □

Claim — The equality $f(0) = 1$ holds.

Proof of the Claim. Substituting $x = 0$, it follows that $f(0)^2 = f(0)$, and hence $f(0) = 0$ or $f(0) = 1$.

On the contrary, suppose $f(0) = 0$ holds. Write $f(x) = x^k g(x)$ where k is a positive integer, and $g(x)$ is a polynomial with real coefficients with $g(0) \neq 0$. The given condition translates to

$$x^k(2x^2)^k g(x)g(2x^2) = (2x^3 + x)^k g(2x^3 + x),$$

which yields

$$(2x^2)^k g(x)g(2x^2) = (2x^2 + 1)^k g(2x^3 + x).$$

Substituting $x = 0$, we obtain $g(0) = 0$, which is impossible. This proves the Claim. □

By the above Claim, the product of the absolute values of the roots of $f(x)$ is equal to 1. By the first Claim, these absolute values are at most 1. It follows that the roots of $f(x)$ are of absolute value 1.

Let α be a root of $f(x)$. Note that $2\alpha^3 + \alpha$ is also a root of $f(x)$, and we have that

$$|\alpha| = |2\alpha^3 + \alpha| = 1,$$

which yields $|2\alpha^2 + 1| = |\alpha| = 1$. This gives

$$(2\alpha^2 + 1)(2\overline{\alpha}^2 + 1) = 1.$$

Combining it with $|\alpha| = 1$, we obtain

$$\alpha^2 + \overline{\alpha}^2 = -2.$$

Since $|\alpha| = 1$, it follows that $\alpha^2 = -1$, and hence, $\alpha = i$ or $\alpha = -i$. So, the polynomial $f(x)$ is equal to $c(x + i)^a(x - i)^b$ for some $c \in \mathbb{C} \setminus \{0\}$, and some nonnegative integers a, b. Since $f(x)$ has real coefficients, it follows that $a = b$, and c lies in $\mathbb{R}$. This gives that $f(x) = c(x^2 + 1)^a$. Using $f(0) = 1$, we obtain $c = 1$, and hence $f(x) = (x^2 + 1)^a$.

Note that if $g(x)$ denotes the polynomial $(x^2 + 1)^k$, where k is a positive integer, then

$$\begin{aligned} g(x)g(2x^2) &= \left((x^2 + 1)((2x^2)^2 + 1)\right)^k \\ &= \left((4x^6 + 4x^4 + x^2 + 1)\right)^k \\ &= \left(((2x^3 + x)^2 + 1)\right)^k \\ &= g(2x^3 + x). \end{aligned}$$

We conclude that the polynomials satisfying the given condition are precisely the constant polynomial $0, 1$, and the polynomials of the form $(x^2 + 1)^k$ for some positive integer k. ■

Theorem 1 (Cauchy)
Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial with complex coefficients such that $a_n \neq 0$ and $n \geq 1$. Then any root $\alpha \in \mathbb{C}$ of $P(x)$ satisfies

$$|\alpha| < 1 + \frac{\max\{|a_{n-1}|, |a_{n-2}|, \ldots, |a_1|, |a_0|\}}{|a_n|}.$$

Moreover, if $k \geq 0$ is an integer such that

$$a_{n-1} = a_{n-2} = \cdots = a_{n-k} = 0,$$

then any root $\alpha \in \mathbb{C}$ of $P(x)$ satisfies

$$|\alpha| \leq 1 + \left(\frac{\max\{|a_{n-k-1}|, |a_{n-k-2}|, \ldots, |a_1|, |a_0|\}}{|a_n|}\right)^{\frac{1}{k+1}}.$$

Proof. Let α be a root of P in $\mathbb{C}$. Write

$$M = \max\{|a_{n-1}|, |a_{n-2}|, \ldots, |a_1|, |a_0|\}.$$

Note that

$$a_n\alpha^n = -(a_{n-1}\alpha^{n-1} + a_{n-2}\alpha^{n-2} + \cdots + a_1\alpha + a_0)$$

holds, which implies

$$|a_n||\alpha|^n \leq M(|\alpha|^{n-1} + |\alpha|^{n-2} + \cdots + |\alpha| + 1).$$

It suffice to consider the case $|\alpha| > 1$, since otherwise the desired inequality follows. Using

$$|\alpha|^n > (|\alpha| - 1)\left(|\alpha|^{n-1} + |\alpha|^{n-2} + \cdots + |\alpha| + 1\right),$$

we obtain

$$(|\alpha| - 1)\,|a_n| < M,$$

which gives the first desired inequality.

Moreover, if $k \geq 0$ is an integer such that

$$a_{n-1} = a_{n-2} = \cdots = a_{n-k} = 0,$$

then we have

$$|a_n||\alpha|^n \leq M(|\alpha|^{n-k-1} + |\alpha|^{n-k-2} + \cdots + |\alpha| + 1).$$

Again, it suffice to consider the case $|\alpha| > 1$. Note that for any real number $r > 1$, writing $r = 1 + s$, we obtain

$$\begin{aligned} r^n &= (1+s)^n \\ &= \sum_{i=0}^{n} \binom{n}{i} s^i \\ &> \sum_{i=k+1}^{n} \binom{n}{i} s^i \\ &\geq \sum_{i=k+1}^{n} \binom{n-k}{i-k} s^i \\ &= s^k \sum_{j=1}^{n-k} \binom{n-k}{j} s^j \\ &= s^k \left((1+s)^{n-k} - 1\right) \\ &= (r-1)^k \left(r^{n-k} - 1\right) \\ &= (r-1)^{k+1} \left(r^{n-k-1} + r^{n-k-2} + \cdots + r + 1\right). \end{aligned}$$

The above implies that

$$|a_n|(|\alpha| - 1)^{k+1} < M,$$

which gives the second desired inequality. □

Exercise 9 (Balkan Math Olympiad 1989 P2, , Yugoslavia)**.** Let $\overline{a_n a_{n-1} \ldots a_1 a_0}$ be the decimal representation of a prime number such that $n > 1$ and $a_n > 1$. Prove that the polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

cannot be expressed as a product of two non-constant polynomials with integer coefficients.

Solution 9. On the contrary, let us assume that

$$P(x) = Q(x)R(x)$$

for some non-constant polynomials $Q(x), R(x)$ with integer coefficients. Since $P(10)$ is a prime, by reordering Q and R if necessary, we may assume without loss of generality that $|Q(10)| = 1$. Using Cauchy's theorem on the size of roots of polynomials (see Theorem 1) and that $a_n \geq 2$, it follows that the roots of $P(x)$ have absolute value less than

$$1 + \frac{9}{2}.$$

The content posted here and at this blog by Evan Chen are quite useful.

Let $r_1, \ldots, r_k$ denote the roots of $Q(x)$ in $\mathbb{C}$, counted with multiplicity. Then we have

$$|Q(10)| = |a| \prod_{i=1}^{k} |10 - r_i|,$$

where a is the leading coefficient of $Q(x)$. Using the triangle inequality, we obtain

$$|Q(10)| \geq |a| \prod_{i=1}^{k} (10 - |r_i|) > |a| \left(10 - \frac{11}{2}\right)^k > 1,$$

which is a contradiction. Hence the polynomial $P(x)$ cannot be expressed as a product of two non-constant polynomials with integer coefficients. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)

[**Goy21**] Rohan Goyal. "Polynomials". Available at https://www.dropbox.com/s/yo31nat6z5ggaue/Polynomials.pdf?dl=0. 2021 (cited p. 2)